Exercise Sheet

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Exercise 1. Show that for any IND-CPA secure PKE scheme $(\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$, its encryption function $\mathsf{Enc}(pk, m)$ must be randomized.

Exercise 2. Given a cyclic group $\mathbb{G} = \langle g \rangle$ of prime order p. Recall the Elgamal Encryption Scheme with message space $\mathcal{M} = \mathbb{G}$ from lecture 2:

$\mathsf{Gen}(1^\lambda)$: $sk \coloneqq x \leftarrow_\$ \mathbb{Z}_p$, $\quad pk \coloneqq h \coloneqq g^x$

$\mathsf{Enc}(pk, m \in \mathbb{G})$: $r \leftarrow_\$ \mathbb{Z}_p$, $\quad c_1 \coloneqq g^r$, $\quad c_2 \coloneqq m \cdot h^r$, $\quad ct \coloneqq (c_1, c_2) \in \mathbb{G} \times \mathbb{G}$

$\mathsf{Dec}(sk, ct)$: Parse $ct = (c_1, c_2)$, output $\dfrac{c_2}{c_1^x}$.

We have proved in class that if the DDH assumption holds in $\mathbb{G}$, then Elgamal is IND-CPA secure. Now we want to prove the converse: Suppose Elgamal is IND-CPA secure, prove that the DDH assumption holds in $\mathbb{G}$.

Exercise 3 (Security Definitions).

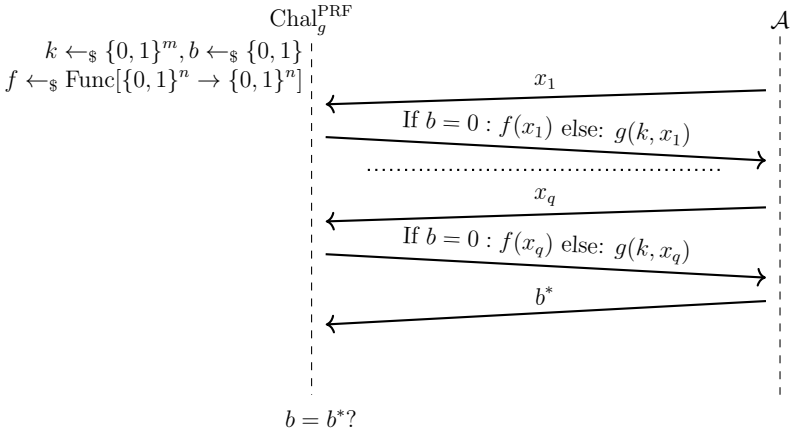
1. Construct a PKE scheme Π_1 that is IND-CPA secure but not IND-CCA1 secure.
2. Construct a PKE scheme Π_2 that is IND-CCA1 secure but not IND-CCA2 secure.

Exercise 4 (PRP-PRF switching lemma). Denote the set $\mathsf{Func}[\{0,1\}^n \to \{0,1\}^n]$ to be all possible functions mapping from $\{0,1\}^n$ to $\{0,1\}^n$, and the set $\mathsf{Perm}[\{0,1\}^n]$ to be all possible permutations on $\{0,1\}^n$.

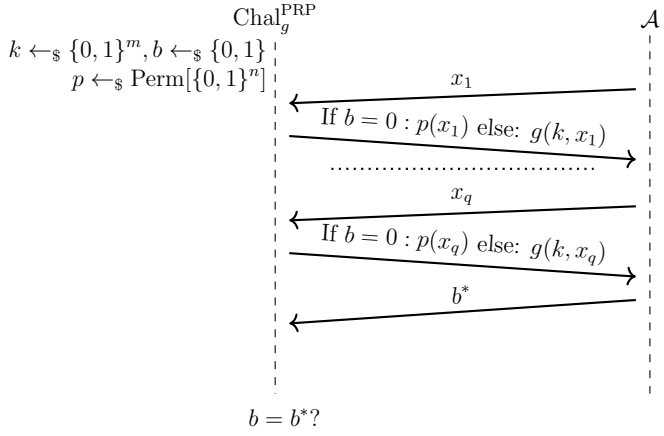
For a block cipher[1] $g : \{0,1\}^m \times \{0,1\}^n \to \{0,1\}^n$, consider the following security games:

In the PRF security game, an adversary $\mathcal{A}$ want to distinguish g with a truly random function f uniformly sampled from $\mathsf{Func}[\{0,1\}^n \to \{0,1\}^n]$. It can make at most q queries to g (or f). The advantage of $\mathcal{A}$ in this game is defined as $\mathbf{Adv}_g^{\mathrm{PRF}}(\mathcal{A})$.

[1] In this exercise, it is sufficient to treat g just as a function that takes two inputs $k \in \{0,1\}^m$ and $x \in \{0,1\}^n$.



In the PRP security game, an adversary $\mathcal{A}$ want to distinguish g with a random permutation p uniformly sampled from $\text{Perm}[\{0,1\}^n]$. It can make at most q queries to g (or p). The advantage of $\mathcal{A}$ in this game is defined as $\mathbf{Adv}_g^{\text{PRP}}(\mathcal{A})$.



Use hybrid argument to prove the following theorem:

Theorem 1. *For any adversary $\mathcal{A}$ making at most q queries to the corresponding challengers, we have*

$$|\mathbf{Adv}_g^{\text{PRP}}(\mathcal{A}) - \mathbf{Adv}_g^{\text{PRF}}(\mathcal{A})| \leq \frac{q^2}{2^n}.$$

(Intuitively, it states that if g is a secure PRP, then it is essentially also a secure PRF.)

Hint.

1. One way to sample a random function (permutation) is to use lazy sampling, e.g. upon a query x that has not been queried before, sample the value of $f(x)$ randomly on the fly (Instead of sampling the whole f at the beginning of the game).

2. What can the hybrids be? Can you establish some relationship on the adversary's output among these hybrids?